

BDA PPTX EXPLANATION

Slide 1: Introduction (Project & Team) "Good morning, everyone. We are Vaishnavi, Sakshi, Bhavya, and Avatar. Today we are presenting our Big Data Recommendation System. Our goal was to build a scalable engine capable of predicting what a user will buy next, similar to the technology used by Amazon and Netflix."

Slide 2: The Dataset & Big Data Challenge "We built this system using a real-world dataset from RetailRocket, which contains over 2.7 million shopping events. The core challenge was scale: processing 100,000 unique users and nearly 110,000 items requires more power than standard local tools can provide."

Slide 3: Apache Spark Architecture "Because regular Python tools would struggle with 2.7 million rows on a single machine, we utilized Apache Spark. Spark solves this by splitting the data across multiple CPU cores, allowing us to process massive chunks of data in parallel, making our system enterprise-ready."

Slide 4: The Core Concept "The foundational idea of our model is: 'You are what you click'. The system is completely blind to product names, categories, or prices. Instead of analyzing the items, it analyzes user behavior to find patterns."

Slide 5: Collaborative Filtering "This concept is known as Collaborative Filtering. Simply put, if User A and User B interact with the exact same sequence of products, the model determines they have identical taste. It will then recommend whatever User B bought to User A, effectively letting users unknowingly filter recommendations for each other."

Slide 6: Implicit Feedback "A major challenge in e-commerce is that users rarely leave 5-star ratings. We solved this using 'Implicit Feedback'. We assigned mathematical weights to user actions: a product view is worth 1.0, adding to cart is 3.0, and completing a transaction is 5.0. We infer their preferences directly from their behavior."

Slide 7: The User-Item Matrix "We transformed these 2.7 million events into a massive table called a User-Item Matrix. This table maps our 100,000 users against all 109,845 items. Because a single user only interacts with a tiny fraction of the catalog, this creates a 'sparse matrix' made up mostly of zeros."

Slide 8: The ALS Algorithm "To process this massive, sparse matrix, we used the Alternating Least Squares (ALS) algorithm. ALS breaks the giant matrix down into two smaller matrices, assigning every user and every item a vector of 20 hidden numbers. You can think of this vector as a unique 'taste fingerprint' for users, and a 'feature fingerprint' for items."

Slide 9: Generating Predictions "To predict how much a user will like an item they haven't seen yet, the model calculates the 'dot product' by multiplying the user's vector and the item's vector. We run this calculation for every unseen item in the catalog, rank the scores, and present the top 5 highest scores as the final recommendations."

Slide 10: Model Accuracy (RMSE) "We evaluated our model on a 20% test set and achieved an RMSE (Root Mean Square Error) of 1.3086. This means our predictions are off by about 1.31 points on average. Because implicit behavioral data includes a lot of natural noise—like window shopping—an RMSE between 1.0 and 2.0 is considered the standard acceptable range in the industry, meaning our model is highly accurate."

Slide 11: Dashboard & Live Demo "We deployed our model into an interactive dashboard. For our demo, we can input Visitor 684514, our most active user, to see how the model confidently generates recommendations based on rich history. If we input an unknown ID like 999, the system triggers a 'Cold Start' warning, demonstrating that we have successfully accounted for new users who lack behavioral history."

Slide 12: Conclusion & Thank You

"Our project successfully demonstrates how big data technologies like Apache Spark and machine learning algorithms like ALS can be combined to understand user behavior at scale. Thank you for your time, we are now open to any questions."